

THE TEST COMPARISON FOR DE MORGAN CUBICAL SETS: REVERSAL, EQUIVARIANCE, AND THE JOIN OF TWO PROJECTIVE SPACES

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ABSTRACT. The cubical-type model structure on De Morgan cubical sets is the semantic home of CCHM cubical type theory; since 2018 it has been folklore that it does not present the homotopy theory of spaces, with the quotient L of the interval by the reversal as the counterexample-in-waiting. We give the first complete proof of the underlying fact — L is not weakly contractible, uniformly in the De Morgan, Kleene and Boolean cube categories — and identify the homotopy type in general form: every quotient of $\square^n$ by a group of signed coordinate permutations acting freely on cells is a $K(H, 1)$. We then close the two known repair routes: any equivariant-style repair that contracts the isotropy quotients — as the cartesian repair of Awodey, Cavallo, Coquand, Riehl and Sattler does by design — necessarily overshoots the test localizer, since the category of elements of a cell-free quotient is a nontrivial classifying space; and no relatively elegant embedding of the De Morgan or Kleene cube category into a Reedy category exists. The folklore witness itself evaporates: the test homotopy type of L is $B\mathbb{Z}/2$, in agreement with the cubical-type answer, so whether the two structures coincide is re-opened. On the re-opened question we prove the upper bound $W_{\text{type}} \subseteq W_{\text{test}}$ on any totally aspherical test cube category, and test its sharpness on all isotropy quotients of the square: cyclic quotients at a fixed stratum contract in both structures, cell-free quotients are $K(H, 1)$ in both, and for the mixed Klein group the category of elements of the quotient is the join $\mathbb{RP}^\infty \star \mathbb{RP}^\infty$, contractible in neither. Every candidate analyzed here has the same weak-contractibility status in the two structures, no automorphism quotient of the square separates the two localizers, and we conjecture the agreement extends to the homotopy types. All finite combinatorial claims are verified by reproducible exhaustive computations.

1. INTRODUCTION

Cubical type theory in the style of Cohen–Coquand–Huber–Mörtberg [8] is interpreted in presheaves on a cube category whose interval carries connections $\wedge, \vee$ and a reversal $\neg$ — the free De Morgan algebra structure. These

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presheaves carry a *cubical-type model structure* (Sattler [16]; Coquand [10] gives an explicit trivial-cofibration/fibration factorization), whose fibrant objects model univalent type theory. They also carry a second Quillen model structure with the same cofibrations: the *test model structure* of Cisinski’s theory [9], whose weak equivalences are detected by the category of elements and which presents the homotopy theory of spaces, because the De Morgan cube category is a strict test category [4, 13]. Whether the two structures coincide — whether CCHM cubical type theory *presents spaces* — is the *test comparison problem*, and it is the subject of this paper.

For the cartesian cube category the answer is negative, by an argument of Sattler and Coquand from June 2018: the quotient of the square by the coordinate swap is not weakly contractible in the cubical-type structure, though it is test-trivial [10, 11]; the equivariant repair of this defect is the recent landmark of Awodey–Cavallo–Coquand–Riehl–Sattler [1]. For the De Morgan cube category the corresponding negative claim — with counterexample the quotient

$$L := \square^1 / (x \sim \neg x)$$

of the representable interval by the reversal — has circulated since the same discussions [11, 15], but no proof has been published: Coquand’s note proves the cartesian case in full and says of the De Morgan case only that the argument “should also be valid” [10, Rem. 3]; the recent literature [1, 7] continues to cite the 2018 lecture [15] for it.

This paper settles what is true and what is not in that folklore, and carries the comparison substantially further. Throughout, $\mathcal{T}$ denotes any of the De Morgan, Kleene, or Boolean interval theories; “the model structure” unqualified means the cubical-type model structure. The results fall into four groups.

I. The reversal quotient, and covering quotients. First the folklore’s true half, in a stronger form than it records (§§3–5):

- *Theorem 4.5* (first published proof; Kleene case new): in all three theories, $L \rightarrow 1$ is not a weak equivalence, although the geometric realization $|L| \cong [0, 1/2]$ is contractible; hence geometric realization is not a Quillen equivalence.
- *Theorem 5.4* (unclaimed in any source, and stated here for all cell-free isotropy quotients): for every subgroup H of signed coordinate permutations acting freely on the cells of $\square^n$, the quotient $\square^n/H$ is a $K(H, 1)$ in the cubical-type model structure; in particular (Corollary 5.5) L is a $K(\mathbb{Z}/2, 1)$.

Everything flows from a one-line *freeness theorem* (Theorem 3.1): the reversal acts without fixed cells on the interval, in every dimension and all three theories — and its calculus, the cell-freeness criterion of Lemma 3.2. For Theorem 4.5 we adapt Coquand’s factorization-invariant method [10, §4–5] from symmetric squares to symmetric lines; two subtleties absent

from the cartesian case appear (connection-degenerate composition directions; and the connection-degenerate line *is* symmetrically contractible, so a finite *reachability census* is essential, not decorative). For Theorem 5.4 we build an honest principal covering at the level of fibrant replacements: freeness makes the quotient map $\square^n \rightarrow \square^n/H$ a fibration by *sheet transport*, the syntactic fibrant replacement commutes with cell-free quotients, and the resulting fibration of fibrant objects has contractible total space and discrete fiber H : *the presheaf quotient computes the homotopy quotient of the action, not its orbit space*.

II. Both repairs fail. Two technologies have repaired or organized such defects on neighbouring cube categories: equivariance [1] and (relative) elegance [6]. Both fail for the reversal, each in the precise sense stated below and each for a structural reason rooted in the freeness theorem (§§6–8):

- *Theorem 6.1:* the automorphism groups of De Morgan and Kleene cubes are the hyperoctahedral groups $G_n = \Sigma_n \ltimes (\mathbb{Z}/2)^n$ — for De Morgan by Birkhoff duality and a centralizer computation, for Kleene by an atom-and-pair analysis of the unmixed point poset.
- *Theorem 7.2:* *any* repair in the style of [1] — any model structure with monomorphism cofibrations in which the isotropy quotients I^k/H become contractible — *overshoots*: by freeness, the category of elements of I^k/H has the homotopy type of BH for every cell-free H , so such a structure contracts objects that the test structure does not, and cannot present spaces through the test comparison — in exact opposition to the cartesian case [1, Thm. 6.3.6]. In one line: *equivariance repairs defects born of fixed cells, and the reversal has none*. We outline the hyperoctahedral transfer of the machinery of [1] and isolate the two modifications it requires, one of which (an orbit-groupoid form of the vertex lemma) simplifies the original; its full verification is deferred to a companion paper, and the overshoot theorem is independent of it (Remark 7.4).
- *Theorem 8.5:* the De Morgan and Kleene cube categories admit no relatively elegant embedding into any Reedy category. The winding-number obstruction of [6, App. A] genuinely dissolves at the level of vertex actions (De Morgan vertex actions are arbitrary Boolean functions); it returns one level deeper, where the required factorization is an equation in free De Morgan algebras: a three-branch forcing pins the factorizing map to antichain fibers, and monotonicity from a connected region into an antichain forces constancy — so the winding number reappears. The Boolean cube category, by contrast, stabilizes.

III. The folklore witness is void; the upper bound. The 2018 record asserts that the cubical-type and test weak equivalences differ on De Morgan cubes, with L as the counterexample-in-waiting. But the test structure does

not see the topological realization; it sees the category of elements, and by freeness (§9)

$N\text{el}(L) \simeq B\mathbb{Z}/2 \simeq$ the cubical-type answer of Corollary 5.5 :

on L the two structures *agree*, and it is the realization that is the outlier (it is not a test functor). With the witness void and both repairs closed, the test comparison problem for De Morgan cubes is genuinely open — and on it we prove the upper bound (§9):

- *Theorem 9.2:* on any totally aspherical test cube category whose cubical-type trivial cofibrations lie in the saturation of the pushout products of endpoint inclusions with monomorphisms,

$$W_{\text{type}} \subseteq W_{\text{test}}.$$

The proof is a one-paragraph localizer argument. On cartesian cubes the inclusion is strict (Coquand’s square); and its contrapositive — test-nontrivial implies type-nontrivial — becomes the engine of the remainder of the paper.

IV. Isotropy quotients and the join. A separating witness for the remaining inclusion must be test-trivial and type-nontrivial. The candidates documented in the literature, and all those considered here, are isotropy quotients $\square^n/H$, so we compute both sides for all subgroups of $\text{Aut}(\square^2) = G_2$ and a representative three-dimensional family (§§10–14):

- *Theorem 10.1* (fixed-stratum retraction): if a cyclic group’s generator fixes a cell, the quotient is contractible in the cubical-type structure — an explicit equivariant interpolation, available because connections let one solve the equivariance constraints around each cycle — hence in both structures. This uniformly contracts the symmetric square, the twisted square, and a $\mathbb{Z}/3$ quotient of the cube; it is exactly what fails for cartesian cubes, where no connections exist and Coquand’s square survives as an obstruction.
- *Theorem 11.1:* the cell-free quotients are $K(H, 1)$ in both structures, matching on the nose (Theorem 5.4 and Lemma 7.1).
- *Theorem 12.2:* for the two remaining subgroups of G_2 — the Klein group $K = \{1, \text{sw}, \text{n}, \text{g}\}$ of the transposition and the double negation, and G_2 itself — no cell is fixed by the whole group, and the category of elements of the quotient is computed exactly: for the Klein group it is the *join*

$$\text{el}(\square^2/K) \simeq B\mathbb{Z}/2 \star B\mathbb{Z}/2 = \mathbb{RP}^\infty \star \mathbb{RP}^\infty,$$

simply connected with $H_3 = \mathbb{Z}/2$: contractible in *neither* structure (type-noncontractibility via Theorem 9.2, and independently by a twisted reachability census in the style of part I).

Every candidate tested — the reversal circle, the symmetric and twisted squares, all cell-free quotients, the Klein and hyperoctahedral quotients, the $\mathbb{Z}/3$ cube — has the same weak-contractibility status in both structures,

with the full homotopy types agreeing on the reversal circle and the cell-free quotients (Table 1); no automorphism quotient of the square separates the two localizers (Corollary 14.1). We conjecture the agreement is general (Conjecture 14.2), identify why the mixed-case type-side computation is genuinely hard (Remark 14.3), and record along the way that the De Morgan and Kleene cube categories are not Eilenberg–Zilber categories (Proposition 8.7): the Kleene midpoint $x \wedge \neg x$ has no (split epi, mono) factorization.

Relation to the literature. Our debt to [10] is total as to method in part I; the contribution there is the adaptation with its two new subtleties, the uniform treatment of the three theories, the previously unexamined Kleene case, the identification of the homotopy types, and machine verification of every finite ingredient. Remark 7.4 outlines an application of the architecture of [1], stating precisely which components transfer unchanged and which need modification; the overshoot phenomenon is the one [1] themselves flag in §1.7.2 (“equivariance is not a catch-all solution”), here given an exact combinatorial diagnosis (cell-freeness) and a proof that is independent of any particular repair. Theorem 8.5 extends [6, App. A] beyond monotone cube maps. The reversal-elimination theorem of Cavallo–Sattler [7] obtains a model of a theory *with* reversals presenting spaces by interpreting over reversal-free cubes; our results delimit what it is they are eliminating: reversal as a *cube-category* symmetry inevitably manufactures $K(\mathbb{Z}/2, 1)$ s from free quotients. For Dedekind cubes (no reversal) the presentation question remains open [18, 6]; the Dedekind and De Morgan questions are now open for dual reasons — there for lack of any candidate argument, here because every candidate witness has been eliminated. The literal-cube combinatorics of free De Morgan and Kleene algebras used throughout is developed in [19, 20], building on the description of free De Morgan algebras of Movsisyan–Aslanyan [14]. All finite combinatorial claims are verified by reproducible exhaustive computations (§15).

2. PRELIMINARIES

2.1. Cube categories and the interval. Following [10, §1], the base category has as objects finite sets of names $I, J, \dots$; a morphism $f : J \rightarrow I$ is a function from I to the free algebra $\mathcal{T}(J)$ of the interval theory $\mathcal{T}$ on the names J , where $\mathcal{T}(J)$ is the free De Morgan algebra $\text{DM}(J)$, the free Kleene algebra $\text{KL}(J)$, or the free Boolean algebra $\text{BA}(J)$. We write $\square_{\text{DM}}, \square_{\text{KL}}$ for the De Morgan and Kleene cube categories, $[n]$ for the object with n names, and $\square^n$ for the representable presheaf. The formal interval is $\mathbb{I}(J) = \mathcal{T}(J)$; recall $|\text{DM}(1)| = 6$, $|\text{DM}(2)| = 168$, $|\text{DM}(3)| = 7,828,354$, $|\text{KL}(3)| = 43,918$.

We use the literal-cube representations [19, 20]: $\text{DM}(J)$ is the lattice of monotone Boolean functions on $\{0, 1\}^{2|J|}$ (coordinates in pairs $(x, \neg x)$), $\text{KL}(J)$ its quotient by the mixed points, $\text{BA}(J)$ the functions on consistent points. The reversal $\neg$ acts on elements by $(\neg t)(\alpha) = 1 - t(\sigma\bar{\alpha})$, where $\bar{\alpha}$ complements all coordinates and σ swaps each pair.

The reversal quotient is

$$L = \square^1 / (t \sim \neg t), \quad L(J) = \mathcal{T}(J) / \sim$$

(colimits of presheaves are computed pointwise). We write $[t]$ for cells of L and $\ell(t)$ when we think of them as reparametrizations of the generating line. Note $[0] = [1]$: L has a single vertex ℓ_0 , and the generator $[x]$ is a loop at it.

2.2. Model structures and localizers. The *cubical-type model structure* on $\widehat{\mathcal{T}}$ has as cofibrations the (decidable) monomorphisms; its fibrations are the maps with a uniform composition structure, equivalently with uniform lifting against the pushout products $\delta_e \hat{\otimes} m$ of an endpoint inclusion $\delta_e : 1 \rightarrow \mathbb{I}$ ($e \in \{0, 1\}$) with a monomorphism m [16, 8]. Its weak equivalences are W_{type} . Recall that in a presheaf category the monomorphisms are the saturation — closure under pushout, transfinite composition and retract — of the *set* of subrepresentable inclusions $\{S \hookrightarrow \square^n\}$; correspondingly the type structure is cofibrantly generated by sets, with generating trivial cofibrations the $\delta_e \hat{\otimes} m$ for m subrepresentable (see Remark 9.3).

Coquand’s note constructs, for any map $\sigma : A \rightarrow B$, an explicit trivial cofibration–fibration factorization $A \rightarrow T \rightarrow B$ where T is the total space of an inductively defined family E over B with constructors inc and $\text{comp}^{k \rightarrow l}(u_k, \psi, u, v)$ ($k \neq l \in \mathbb{I}(J)$, $\psi \neq 1$), and restriction maps given by: reindexing when $kg \neq lg$ and $\psi g \neq 1$; $u_k g$ when $kg = lg$; $u_{[l]g}$ when $\psi g = 1$ [10, §3]. A map is a weak equivalence iff in (any) such factorization the fibration part is a trivial fibration; trivial fibrations have sections. All of this is verbatim uniform in $\mathcal{T}$. We write $T(X)$ for the total space of the factorization of $X \rightarrow 1$: the *syntactic fibrant replacement*.

The *test model structure* has the same cofibrations and weak equivalences $W_{\text{test}} = \{f : \text{Nel}(f) \text{ a simplicial weak equivalence}\}$, where el is the category of elements; it exists and presents spaces because the three cube categories are strict — indeed totally aspherical — test categories [9, 13, 4].

An *A-localizer* [9, §1.4] is a class W of maps of $\widehat{A}$ containing the maps with the right lifting property against all monomorphisms, satisfying two-out-of-three, and whose monomorphic members are closed under pushout, transfinite composition, and retract. W_{test} is an *A-localizer*; W_{min} denotes the minimal one.

Proposition 2.1. *On any of the three cube categories, W_{type} is an A-localizer; consequently $W_{\text{min}} \subseteq W_{\text{type}}$.*

Proof. All the axioms are theorems of model-category theory for a model structure whose cofibrations are the monomorphisms; the minimal localizer is contained in every localizer. $\square$

2.3. Automorphisms; notation for subgroups. The signed symmetric group $G_n = \Sigma_n \times (\mathbb{Z}/2)^n$ acts on cells $(a_1, \dots, a_n) \in \mathcal{T}(J)^n$ of $\square^n$ by permuting the tuple and applying $\neg$ componentwise — no substitution of

variables is involved. That these are *all* the automorphisms of $[n]$ is Theorem 6.1 below. For $n = 2$ we write $\text{sw}(a, b) = (b, a)$, $\text{n}(a, b) = (\neg a, \neg b)$, $\text{g} = \text{swn} : (a, b) \mapsto (\neg b, \neg a)$, and $\text{n}_1(a, b) = (\neg a, b)$; $G_2 = \langle \text{sw}, \text{n}_1 \rangle$ is dihedral of order 8 and $K = \{1, \text{sw}, \text{n}, \text{g}\}$ is a Klein four-group.

3. THE FREENESS THEOREM AND THE CELL-FREENESS CRITERION

Theorem 3.1 (freeness). *In all three theories and for every J , no $t \in \mathcal{T}(J)$ satisfies $t = \neg t$. Hence the reversal action on the cells of $\square^1$ is free, and every cell of L has exactly two lifts $\{t, \neg t\}$.*

Proof. Let α_0 be the point of the literal cube with every pair equal to $(0, 1)$ — the consistent point of the all-zeros assignment, present in all three representations. Then $\sigma \bar{\alpha}_0 = \alpha_0$: complementing gives all pairs $(1, 0)$, and swapping each pair returns $(0, 1)$. So $t = \neg t$ would give $t(\alpha_0) = 1 - t(\sigma \bar{\alpha}_0) = 1 - t(\alpha_0)$, impossible for a Boolean value. $\square$

(Machine census: there are 0 self-dual elements among all 7,828,354 elements of $\text{DM}(3)$, all 43,918 of $\text{KL}(3)$, and below; see §15.)

Lemma 3.2 (cell-freeness criterion). *An element $h = (\pi, \varepsilon) \in G_n$ fixes some cell of $\square^n$ if and only if every cycle of π has even negation weight $\sum_{i \in \text{cycle}} \varepsilon_i$. Consequently a subgroup $H \leq G_n$ acts freely on cells (“is cell-free”) iff every nontrivial element of H has a cycle of odd negation weight.*

Proof. The action permutes the components of a tuple and applies $\neg$; no variables are substituted. A fixed tuple satisfies, around each cycle $(i_1 \cdots i_k)$ of π , a chain of equations $a_{i_{j+1}} = \neg^{\varepsilon} a_{i_j}$, which compose to $a_{i_1} = \neg^w a_{i_1}$ with w the cycle’s weight. For odd w this contradicts Theorem 3.1; for even w any choice of a_{i_1} solves the cycle, and the cycles are independent. $\square$

Examples in G_2 : sw and g fix cells — the diagonal (a, a) and the antidiagonal $(a, \neg a)$ respectively — while n , n_1 , n_2 and $\text{n}_1 \text{sw}$ are cell-free. Note that $\text{g} = (\text{sw}, (1, 1))$ contains reversal components yet is not cell-free: the criterion is the parity of the cycle weights, not the presence of reversals.

4. THE REVERSAL QUOTIENT IS NOT WEAKLY CONTRACTIBLE

Definition 4.1. A square $c(x, z)$ *symmetrically connects* lines $a(x)$ and $b(x)$ if $c(x, 0) = a$, $c(x, 1) = b$ (or vice versa) and $c(\neg x, z) = c(x, z)$. Write $P'(A)$ for the property: every reversal-symmetric line $a(x) = a(\neg x)$ of A is connected to a constant line by a finite chain of symmetrically connecting squares. (In a quotient like L , symmetry of $[t]$ means $t \circ \sigma_x \in \{t, \neg t\}$.)

Lemma 4.2. *If $u : A \rightarrow B$ has a section then $P'(A)$ implies $P'(B)$.*

Proof. As in [10, Lem. 4.1]: push the symmetric line along the section, connect there, map the chain back. $\square$

Proposition 4.3. *For every $\sigma : A \rightarrow B$ with factorization $A \rightarrow T \rightarrow B$ as in §2.2, $P'(A)$ implies $P'(T)$.*

Proof. A symmetric line of T over $v[l]$ is either $(\sigma a, \text{inc } a)$ with a a symmetric line of A — handled by $P'(A)$ — or $(v(x, l), \text{comp}^{k \rightarrow l}(a, \psi, u, \langle z \rangle v(x, z)))$ with $k \neq l \in \mathbb{I}(x)$. Two remarks before the induction. First, the substitution $g = (x \mapsto \neg x)$ is an *automorphism* of $\mathcal{T}(J)$, so the degenerate restriction clauses cannot fire under it: $kg = lg$ iff $k = l$, and $\psi g = 1$ iff $\psi = 1$. Symmetry of the cell is therefore an equality of **comp**-constructors, forcing $k \circ \sigma_x = k$, $l \circ \sigma_x = l$, a symmetric, ψ , u , v symmetric. Second, the fixed elements of $\mathbb{I}(x)$ under $x \mapsto \neg x$ are $\{0, 1\}$ in the Boolean theory but

$$\{0, 1, x \wedge \neg x, x \vee \neg x\}$$

in the De Morgan and Kleene theories: the composition may enter or exit through a connection-degenerate direction, a case with no cartesian analogue. The induction is nevertheless uniform: with a fresh name t , the square

$$c(x, t) = (v(x, t), \text{comp}^{k \rightarrow t}(a, \psi, u, \langle z \rangle v(x, z)))$$

is symmetric (all its data is), equals the given line at $t = l$, and at $t = k$ equals $(v[k], a)$ by the $kg = lg$ restriction clause. Since a has smaller inductive height and is symmetric, induction reaches an **inc**-cell and $P'(A)$ concludes. $\square$

Proposition 4.4. $P'(L)$ fails, in all three theories.

Proof. The symmetric lines of L are exactly its three (Boolean: two) lines: the generator $[x]$, the connection-degenerate $[x \wedge \neg x] = [x \vee \neg x]$ (absent in the Boolean case), and the constant $[0] = [1]$: every element of $\mathcal{T}(x)$ is orbit-symmetric. A symmetrically connecting square of L is an orbit $[t]$, $t \in \mathcal{T}(x, z)$, with $t \circ \sigma_x \in \{t, \neg t\}$ — a *finite* set, and the faces $z = 0, 1$ of its members are computable. The exhaustive census (§15): the symmetric squares number 60 (De Morgan), 32 (Kleene), 8 (Boolean), and the induced graph on symmetric lines is, in each theory,

$$\underbrace{[x]}_{\text{isolated}} \circlearrowleft [x \wedge \neg x] \longleftrightarrow \text{const} \circlearrowright,$$

i.e. every symmetric square with one face $[x]$ has both faces $[x]$. Two features deserve note. The degenerate line *is* symmetrically contractible — $t = (x \wedge \neg x) \wedge \neg z$ is exactly symmetric and connects it to 0 — so the isolation of $[x]$ is essential and not a foregone conclusion; and the connection trick that kills the cartesian symmetric-square invariant [10, Rem. 1] produces $\ell(x \vee z)$, which is *not* symmetric ($x \vee z \neq \neg x \vee z$ and $\neq \neg(x \vee z)$ in the free algebras), in accordance with the census. Hence no chain connects $[x]$ to a constant. $\square$

Theorem 4.5. *In all three theories $L \rightarrow 1$ is not a weak equivalence, although $|L| \cong [0, 1]/(s \sim 1 - s) \cong [0, 1/2]$ is contractible; hence geometric realization is not a Quillen equivalence for the cubical-type model structure.*

Proof. If $L \rightarrow 1$ were a weak equivalence then so would be the vertex $1 \rightarrow L$ (two-out-of-three); factor it as $1 \rightarrow T \rightarrow L$; the fibration $T \rightarrow L$ is then trivial, hence has a section. $P'(1)$ holds trivially, Proposition 4.3 gives $P'(T)$, and Lemma 4.2 gives $P'(L)$ — contradicting Proposition 4.4. Since cofibrations are monomorphisms, L is cofibrant, and a left Quillen equivalence reflects weak contractibility of cofibrant objects [12, 1.3.16]; realization fails to do so. $\square$

(What Theorem 4.5 does *not* show is that the cubical-type structure differs from the *test* structure — the folklore’s second half. That question is taken up in §9, where it turns out the folklore is wrong about it.)

5. COVERING QUOTIENTS ARE EILENBERG–MACLANE OBJECTS

In this section $H \leq G_n$ is any cell-free subgroup (Lemma 3.2), in any of the three theories; the reader interested only in the reversal quotient may take $n = 1$ and $H = \langle \neg \rangle \cong \mathbb{Z}/2$ throughout. We show that $\square^n/H$ is a $K(H, 1)$ in the cubical-type model structure, by exhibiting an honest principal covering at the level of fibrant replacements. The section has four steps: sheet transport, syntactic commutation, the discrete-fiber lemma, and the assembly.

Theorem 5.1 (sheet transport). *For cell-free $H \leq G_n$, the quotient map $p : \square^n \rightarrow \square^n/H$ is a uniform fibration.*

Proof. In dependent form, the fiber over a cell $v \in (\square^n/H)(J)$ is $E(v) = Ht_0$, the H -orbit of any lift t_0 — an H -torsor, by freeness. Given composition data — $v \in (\square^n/H)(J^+)$, $k \neq l \in \mathbb{I}(J)$, a lift $u_k \in E(v[k])$, $\psi \neq 1$, and a compatible partial family $u_f \in E(vf)$ for $f : K \rightarrow J^+$ with $\psi \mathbf{p}f = 1$ — there is a *unique* lift $T \in \square^n(J^+)$ of v with $T[k] = u_k$: the lifts of v form an orbit HT_0 , and $h \mapsto (hT_0)[k] = h(T_0[k])$ is injective by freeness at level J . Define

$$\mathbf{comp}^{k \rightarrow l}(u_k, \psi, u) := T[l].$$

Tube forcing: for $g : K \rightarrow J$ with $\psi g = 1$, the tube cell u_{g^+} (part of the data, as $\psi \mathbf{p}g^+ = 1$) satisfies $u_{g^+}[kg] = u_{[k]g} = u_k g = T[k]g = (Tg^+)[kg]$, using $[k] \circ g = g^+ \circ [kg]$; by freeness at level K^+ , agreeing with Tg^+ at one face forces $u_{g^+} = Tg^+$, whence $u_{[l]g} = u_{g^+}[lg] = (Tg^+)[lg] = T[l]g$: the composition agrees with the prescribed data on ψ . *Uniformity:* for g with $kg \neq lg$, $\psi g \neq 1$, the unique lift of vg^+ through $u_k g$ is Tg^+ , so $\mathbf{comp}(\dots)g = T[l]g = \mathbf{comp}(u_k g, \psi g, u_{g^+})$ — naturality of the construction. Finally $\mathbf{comp}^{k \rightarrow k} = T[k] = u_k$. $\square$

Lemma 5.2 (syntactic commutation). *For cell-free H acting on $X = \square^n$, the H -action extends to the syntactic fibrant replacement $T(X)$ by functoriality, acts freely on its cells, and the canonical comparison is an isomorphism*

$$T(X)/H \cong T(X/H).$$

Proof. The cells of $T(X)$ are the inductively generated terms $\text{inc } a$ (a a cell of X) and $\text{comp}^{k \rightarrow l}(a, \psi, u, v)$, modulo the restriction clauses of §2.2; the action is $h(\text{inc } a) = \text{inc}(ha)$ and $h \text{comp}^{k \rightarrow l}(a, \psi, u, v) = \text{comp}^{k \rightarrow l}(ha, \psi, hu, v)$ — the interval data k, l, ψ and the base v (a cell of the terminal object) are untouched. Since the restriction clauses test interval data only ($kg = lg$; $\psi g = 1$), the action respects them.

Face-datum chains. Each comp -cell carries a distinguished subterm, its face datum a (the cell prescribed at the face k); iterating, every term has a strictly height-decreasing chain of face data ending at an inc -cell, its *base*.

Freeness on $T(X)$ -cells. The action preserves constructors, so a nontrivial h fixing a term fixes its face datum, hence by induction its base $\text{inc } a$, hence the cell a of X — impossible, as H is cell-free on X .

The comparison. Functoriality of $T(-)$ along the quotient map gives an H -invariant map $T(X) \rightarrow T(X/H)$, hence $\varphi : T(X)/H \rightarrow T(X/H)$. We prove, by a single induction on constructor height k (simultaneously over all levels J), the two statements:

- (U_k) any two cells of $T(X)$ of height $\leq k$ with the same image in $T(X/H)$ differ by a unique $h \in H$;
- (S_k) every cell of $T(X/H)$ of height $\leq k$ has a preimage.

All appeals below are to the statements at strictly smaller height, so there is no circularity. *Base:* at height 0 both are the definition of X/H together with cell-freeness of H on X (uniqueness of h).

Step for (U_k). Let t, t' have height $\leq k$ and equal images. The restriction clauses test interval data, which the action and φ leave unchanged, so t, t' are simultaneously degenerate — reducing to smaller height — or both non-degenerate comp -constructors with the same interval data (k, l, ψ) . Their face data have height $< k$ and equal images, so by (U_{k-1}) they differ by a unique h ; replacing t by ht , we may assume the face data are equal and must show $t = t'$. Corresponding tube cells have height $< k$, equal images, and equal restrictions at the k -face (both restrict there to faces of the common face datum); by (U_{k-1}) they differ by some $h' \in H$, and h' fixes the common k -face restriction, so $h' = 1$ by freeness on $T(X)$ -cells. Hence the tubes agree and $t = t'$. The h is unique: two such would produce a nontrivial element fixing t' , contradicting freeness.

Step for (S_k). An $\text{inc}[a]$ -cell lifts to $\text{inc } a$. For a comp -cell of height k , lift the face datum by (S_{k-1}); each tube cell restricts at the k -face to a face of the face datum, already lifted, and by (U_{k-1}) together with freeness (as in the previous step) there is *exactly one* lift of each tube cell agreeing with it at the k -face. The lifted family is compatible and natural in restrictions: corresponding restrictions have equal images and agree at the k -face, so they are equal by the same uniqueness. The lifted data assembles to a comp -cell lifting the given one. $\square$

Lemma 5.3 (discrete fibers and the fiber sequence). *Let R be a fibrant, weakly contractible cubical set carrying an action of a finite group H that*

is free on cells. Then R/H is fibrant; $R \rightarrow R/H$ is a uniform fibration; its fiber over any vertex is the constant presheaf on an H -torsor, which is fibrant; and, pointed at any vertex,

$$\Omega(R/H) \simeq H \text{ (discrete),} \quad \pi_1(R/H) \cong H, \quad \pi_n(R/H) = 0 \text{ (} n \geq 2 \text{)}$$

in the homotopy category of the cubical-type model structure, the group structure on π_1 given by the monodromy of the covering. That is, R/H is a $K(H, 1)$.

Proof. The sheet-transport argument of Theorem 5.1 used only freeness of the action on cells, so it applies verbatim: $R \rightarrow R/H$ is a uniform fibration. R/H is fibrant: an open box in R/H lifts to R — uniquely through any single one of its cells, and open-box domains are connected (unions of faces glued along shared subfaces), so the lift of one cell propagates to the whole box by tube forcing — and is filled in R , and the filler projects. The fiber over a vertex b : its cells over J are the lifts of the degeneracies of b , i.e. the H -orbit of one vertex of R together with its degeneracies — the constant presheaf on an H -torsor. A constant presheaf on a finite set is fibrant: box domains are connected, so partial data is constant, and its value fills the box.

For the loop-space identification we make the model-categorical steps explicit. (a) We work in the category of pointed cubical sets, pointed at a chosen vertex b_0 of R and its images; every object of the model structure is cofibrant, and R , R/H , the fiber F and the point are fibrant, so the pointed homotopy category with its loop functor and fiber sequences is available [12, §§6.1–6.2]. (b) The fiber-sequence machinery of [12, §6.2] applies directly to a fibration of fibrant objects with its *strict* fiber — no properness hypotheses enter — and extends to the left to a sequence $\Omega(R/H) \rightarrow F \rightarrow R \rightarrow R/H$. (c) Since R is weakly contractible, for every Z the exact sequence of pointed sets $[Z, \Omega R] \rightarrow [Z, \Omega(R/H)] \rightarrow [Z, F] \rightarrow [Z, R]$ has trivial outer terms, so $[Z, \Omega(R/H)] \rightarrow [Z, F]$ is a bijection, natural in Z ; by Yoneda in the homotopy category, $\Omega(R/H) \simeq F$. In particular $\pi_n(R/H) \cong \pi_{n-1}(F) = 0$ for $n \geq 2$, F being discrete. (d) Connectivity and the group structure: R is connected and $R \rightarrow R/H$ is surjective, so R/H is connected. Lifts of paths and of homotopies through a prescribed vertex exist and are unique by sheet transport, so reading off the endpoint of the lift at a chosen basepoint lift r_0 defines the monodromy map $\pi_1(R/H, b) \rightarrow H$, an anti-homomorphism for the deck action. It is *surjective*: R is fibrant with $\pi_0(R) = *$, so for each $h \in H$ the vertices r_0 and hr_0 are joined by a finite edge path of 1-cells; its projection is an edge-path loop at b whose monodromy — the composite of the edgewise transports — is h . It is *injective*: a loop with trivial monodromy lifts to a loop at r_0 , which is null-homotopic because R is fibrant and weakly contractible, and the projection of the null-homotopy exhibits the original loop as null-homotopic. Composing with inversion gives $\pi_1(R/H) \cong H$ (every group is isomorphic to its opposite). $\square$

Theorem 5.4. *For every cell-free $H \leq G_n$, in all three theories, $\square^n/H$ is a $K(H, 1)$ in the cubical-type model structure.*

Proof. Let $R = T(\square^n)$: fibrant, and weakly contractible because $\square^n \rightarrow T(\square^n)$ is a trivial cofibration and $\square^n$ is contractible (the connection homotopy $\vec{x} \mapsto \vec{x} \wedge \neg z$ contracts it to the origin). By Lemma 5.2, $T(\square^n)/H \cong T(\square^n/H)$, which is the fibrant replacement of $\square^n/H$; by Lemma 5.3 it is a $K(H, 1)$. $\square$

Corollary 5.5. *In all three theories, L is a $K(\mathbb{Z}/2, 1)$. Concretely, the generator loop $[x]$ lifts to the path x from 0 to 1, crossing sheets: its monodromy is the nontrivial element of $\mathbb{Z}/2$ — this re-proves Theorem 4.5 independently of §4 — while $[x] \cdot [x]$ lifts to a genuine loop and is null-homotopic: $\pi_1 = \mathbb{Z}/2$ exactly.*

Remark 5.6. The two proofs of non-contractibility are one: the reachability census of Proposition 4.4 separates the symmetric lines precisely by the parity of their monodromy — $[x]$ lifts $0 \rightarrow 1$ (odd), while $[x \wedge \neg x]$ and the constants lift to loops (even) — and symmetric squares preserve the parity. The census is the combinatorial shadow of $\pi_1 = \mathbb{Z}/2$.

Remark 5.7. Theorem 5.4 says the presheaf quotient computes the *homotopy* quotient $(\square^n)_{hH} \simeq BH$ rather than the orbit space — because the action is free on cells even when its topological realization has fixed points (for the reversal, at $\frac{1}{2}$). This is precisely the mechanism that equivariant fibrations neutralize on the cartesian side [1], where the symmetric quotients I^n/H play the role of L ; whether an equivariant theory repairs the De Morgan model is the subject of §§6–7.

6. THE AUTOMORPHISM GROUPS ARE HYPEROCTAHEDRAL

Theorem 6.1. *In both the De Morgan and the Kleene cube categories, $\text{Aut}([n]) \cong G_n = \Sigma_n \ltimes (\mathbb{Z}/2)^n$, generated by the coordinate permutations and the reversals $x_i \mapsto \neg x_i$.*

Proof. An automorphism of $[n]$ is an automorphism of the free algebra.

De Morgan case: by Birkhoff duality, lattice automorphisms of the free De Morgan algebra correspond to order-automorphisms of the literal cube $\{0, 1\}^{2n}$, which are exactly the coordinate permutations Σ_{2n} ; compatibility with the involution $\neg$ (complement composed with the pair swap τ) amounts, since complementation is central, to centralizing τ — a product of n disjoint transpositions, whose centralizer in Σ_{2n} is $\Sigma_n \ltimes (\mathbb{Z}/2)^n$: permute the pairs, swap within a pair, the latter realizing $x_i \mapsto \neg x_i$.

Kleene case: the dual object is the unmixed point poset $U_n \subseteq \{0, 1\}^{2n}$, the literal points having no pair $(1, 1)$ together with another pair $(0, 0)$ — exactly the points where the Kleene inequality $x \wedge \neg x \leq y \vee \neg y$ holds pointwise. Thus $U_n = A \cup B$ with A the points with no $(0, 0)$ -pair and B those with no $(1, 1)$ -pair, glued along the Boolean points, and $|U_n| = 2 \cdot 3^n - 2^n$; the constraint

couples the pairs, so U_n is *not* a product of one-pair posets. (For $n = 1$ nothing is excluded, U_1 is the full literal square, and indeed $\text{KL}(1) = \text{DM}(1)$ has order 6.) $\text{KL}(n)$ is the lattice of monotone Boolean functions on U_n , so by Birkhoff duality its lattice automorphisms are the poset automorphisms of U_n , which we now classify for $n \geq 2$ (for $n = 1$ the automorphism group of the square is $\mathbb{Z}/2$, realized by the reversal).

U_n has a unique minimum (all pairs $(0,0)$) and exactly $2n$ atoms, the points with a single bit set; an automorphism permutes the atoms. *Two atoms lie in the same literal pair iff they have no least upper bound*: for atoms in distinct pairs, the bitwise join is again unmixed and is the least upper bound; for the two atoms of one pair, every upper bound has that pair equal to $(1,1)$, hence no $(0,0)$ -pair, and the minimal such points — that pair $(1,1)$, every other pair $(0,1)$ or $(1,0)$ — are $2^{n-1} \geq 2$ pairwise incomparable elements. So an automorphism permutes the n pairs and, within each pair, the two literals: on atoms it agrees with a unique signed permutation $g \in G_n$. A point of U_n is determined by the set of atoms below it, and an order-automorphism preserves this set, so the automorphism is determined by its action on atoms and equals g . Conversely every signed permutation preserves the defining condition of U_n (it permutes pairs and preserves the properties “some pair is $(1,1)$ ” and “some pair is $(0,0)$ ”), so $\text{Aut}(U_n) = G_n$. Every signed permutation commutes with the Kleene involution, so the algebra automorphism group is all of G_n . (Machine check at $n = 2$: $|U_2| = 14$; the monotone functions on U_2 number $84 = |\text{KL}(2)|$, confirming the representation; the poset automorphisms number exactly 8, all commuting with the involution; and the same-pair atom criterion holds — two minimal upper bounds for same-pair atoms, a least upper bound otherwise; §15.) $\square$

7. EQUIVARIANT REPAIRS OVERSHOOT

On cartesian cubes, the repair for the analogous defects is *equivariance*: requiring the uniform filling structures to be equivariant under the automorphisms of the filling cube enlarges the weak equivalences until the quotients I^k/H become contractible, and [1] prove the resulting model structure coincides with the test structure and presents spaces. Theorem 6.1 identifies the corresponding symmetries here; this section shows that *any* repair of this kind necessarily overshoots the test structure on De Morgan and Kleene cubes.

Lemma 7.1. *Let $H \leq G_k$ be cell-free (Lemma 3.2). Then $N\text{el}(I^k/H) \simeq BH$; in particular $N\text{el}(L) \simeq B\mathbb{Z}/2$.*

Proof. First, $\text{el}(I^k/H) \cong \text{el}(I^k)/H$: explicitly, the cells of I^k/H are the H -orbits of cells of I^k , so the objects of $\text{el}(I^k/H)$ are pairs (J, Hf) , and a morphism $(J, Hf) \rightarrow (J', Hg)$ is a cube map $u : J \rightarrow J'$ with $g \circ u \in Hf$; the orbit category $\text{el}(I^k)/H$ — objects the H -orbits of objects of the slice $\text{el}(I^k)$, morphisms the H -orbits of morphisms, whose underlying cube maps

are well defined because the action leaves them unchanged — has exactly the same description. The slice $\text{el}(I^k)$ has a terminal object, so its nerve is contractible; the action is free on objects, hence free on all composable strings, so the quotient functor is a covering of categories and $N(\text{el}(I^k)/H) = N(\text{el}(I^k))/H$ — the quotient of a contractible free H -complex: a BH . $\square$

Theorem 7.2 (any repair overshoots). *Let $\mathcal{W}_{\text{eq}}$ be the class of weak equivalences of any model structure on De Morgan or Kleene cubical sets whose cofibrations are the monomorphisms, and suppose $\square^k/H \rightarrow 1 \in \mathcal{W}_{\text{eq}}$ for some k and some nontrivial cell-free $H \leq G_k$ — as an equivariant-style repair does whenever it makes the vertex inclusions $1 \rightarrow I^k/H$ trivial cofibrations, which is what such repairs are designed to do. Then $\mathcal{W}_{\text{eq}} \not\subseteq \mathcal{W}_{\text{test}}$: the structure does not coincide with the test model structure and cannot present spaces through the test comparison — in exact opposition to the cartesian case [1, Thm. 6.3.6].*

Proof. $N \text{el}(\square^k/H) \simeq BH \not\simeq *$ by Lemma 7.1, so $\square^k/H \rightarrow 1 \notin \mathcal{W}_{\text{test}}$. $\square$

Remark 7.3. The diagnosis, in one line: *equivariance repairs defects born of fixed cells, and the reversal has none.* On cartesian cubes the symmetric quotients I^n/H are test-contractible — their element categories are glued along the fixed diagonal cells — so enlarging the localizer to contract them moves the cubical-type structure *toward* the test structure, and [1] shows it lands exactly on it. On De Morgan cubes the freeness theorem makes the cell-free quotients test-nontrivial (BH 's), so the same enlargement moves *past* the test structure. This realizes, with a proof, the caution of [1, §1.7.2] that “equivariance is not a catch-all solution”: the BCH cube category was their suspected example; the De Morgan cube category is a proven one.

Remark 7.4 (the transfer of the cartesian machinery). Theorem 7.2 is repair-agnostic; the repair it targets is nevertheless expected to exist. In outline, the equivariant machinery of [1] transfers to De Morgan and Kleene cubical sets over the hyperoctahedral groupoid of Theorem 6.1, with two modifications: the species-level interval has no global points (the reversal moves every vertex — Theorem 3.1 at the vertex level), so the species layer is demoted to a generation device and the model-structure arguments run at the cubical-set level; and the vertex lemma [1, Lem. 5.2.5], whose proof requires the automorphism to fix the vertex, is replaced by a colimit over the orbit groupoid of the action, which needs no fixing hypothesis and makes every vertex inclusion $1 \rightarrow I^k/H$ a trivial cofibration. A complete verification parallels [1, §§2–5] at length and is deferred to a companion paper; nothing in the present paper depends on it — by Theorem 7.2, any construction of this kind overshoots, whatever its details.

8. NO RELATIVELY ELEGANT EMBEDDING

We now close the second route. Recall from [6, App. A] the crown posets Cr_m (cyclic fences of $2m$ elements), the embeddings $c_m : \text{Cr}_m \hookrightarrow [1]^m$, the

wrapping maps $f_a : \text{Cr}_{am} \rightarrow \text{Cr}_m$ of winding number a , and the extensions $\bar{f} : [1]^{am} \rightarrow [1]^m$ sending $c_{am}(i) \mapsto c_m(f(i))$, $\perp \mapsto \perp$, and everything else to $\top$; the three branches have disjoint values, so the defining square is a pullback. If some $i : \mathcal{C} \rightarrow R$ were relatively elegant, principal sieves on each object of $\mathcal{C}$ would carry a strictly monotone degree [6, Lem. A.5] — the lemma’s fullness hypothesis is part of the definition of relative elegance, so it applies to any $\mathcal{C}$ — and the chain $\langle \bar{f}_2 \rangle \supseteq \langle \bar{f}_4 \rangle \supseteq \cdots$ of principal sieves on $[1]^n$ would stabilize: some $\bar{f}_a = \bar{f}_b \circ g$ with $a < b$ and $g : [1]^{an} \rightarrow [1]^{bn}$ a map of $\mathcal{C}$.

Remark 8.1 (anatomy of the proof). The following are imported from [6, App. A] unchanged: the crown posets, the embeddings c_m , the wrapping maps f_a and their extensions $\bar{f}_a$, the disjointness of the three branches — which makes the defining square a pullback and gives $\bar{f}_b^{-1}(\text{crown values}) = c_{bn}(\text{Cr}_{bn})$ exactly — and the sieve-degree consequence of relative elegance [6, Lem. A.5], which reduces the theorem to the nonexistence of a factorization $\bar{f}_b \circ g = \bar{f}_a$ in the cube category. What fails here is the original winding argument: it lives at the level of vertices, and De Morgan vertex actions are arbitrary Boolean functions, so a set-level section provides a vertex-level factorization. What is new is the analysis of the factorization equation at the level of free algebras: the pullback property forces the factorizing map onto the crown region and into the wrapping fibers, the fibers are antichains (Lemma 8.2), the region is connected (Lemma 8.3), monotonicity then collapses the negative-literal freedom (Lemma 8.4), and the winding obstruction returns.

Three elementary facts about the combinatorics, kept self-contained:

Lemma 8.2. *In a cyclic fence, two distinct elements are comparable iff they are cyclically adjacent. Consequently each fiber $f_b^{-1}(y)$ of the wrapping map — b elements spaced $2m$ apart in Cr_{bm} — is an antichain for $m \geq 1$, and distinct fibers are disjoint antichains.*

Proof. A cyclic fence alternates minimal and maximal elements around a cycle, so the only comparabilities are between cyclic neighbours; elements at cyclic distance $2m \geq 2$ are therefore incomparable. $\square$

Lemma 8.3. *The crown region — the product of the crown subposet with the Boolean cube of negative literals — is connected as a poset, and so is its restriction to the unmixed (Kleene) points.*

Proof. De Morgan: the crown is connected (a cyclic zigzag), the Boolean cube is connected (flip one coordinate at a time through comparable pairs), and a product of connected posets is connected.

Kleene: the region is the intersection with the unmixed points and is no longer a product (the unmixedness constraint couples the pairs; cf. Theorem 6.1), so we connect its points by explicit zigzags of single-bit flips, each flip a comparability. First, every point of the region is connected within it to

the Boolean point over the same crown vertex, with each pair (c_i, ν_i) moved to $(c_i, \neg c_i)$: an unmixed point has no $(1, 1)$ -pair or no $(0, 0)$ -pair; in the first case, moving its non-Boolean pairs — all of the form $(0, 0)$ — to $(0, 1)$ one bit at a time creates no $(1, 1)$ -pair, so every intermediate point is unmixed; the second case is dual. Second, the Boolean points over crown-adjacent vertices are connected: the two vertices differ in one positive coordinate, flipping it produces a single non-Boolean pair — $(1, 1)$ or $(0, 0)$ — while all other pairs are Boolean, so the intermediate point is unmixed either way; flipping the partner negative bit restores a Boolean point. Since the crown is connected, so is the region. $\square$

Lemma 8.4. *A monotone map from a connected poset to an antichain is constant.*

Proof. Comparable elements map to comparable elements, which in an antichain are equal; constancy propagates along the zigzags of comparabilities that connect the poset. $\square$

Theorem 8.5. *For $\mathcal{C} \in \{\square_{\text{DM}}, \square_{\text{KL}}\}$, no such g exists; hence the De Morgan and Kleene cube categories admit no relatively elegant embedding into any Reedy category.*

Proof. A first phenomenon, absent from the Dedekind case: at the level of *vertex actions* the factorization exists. De Morgan cube maps have arbitrary Boolean vertex actions (disjunctive normal forms over literals are functionally complete), the three branches of $\bar{f}_b$ are disjoint, and a set-level section of f_b over the crown extended by $\perp \mapsto \perp$, else $\top$, satisfies the vertex-level equation — no monotonicity, hence no winding number, obstructs it. The obstruction returns at the free-algebra level, where the required equation $\bar{f}_b \circ g = \bar{f}_a$ holds in free algebras, i.e. as monotone functions on the literal cube. Write G for the induced monotone map on literal points and pos for the projection to positive literals. Since $\bar{f}_b$ is negation-free, the composite's literal function is $\bar{f}_b$'s vertex action applied to G ; since $\bar{f}_b^{-1}(\text{crown values}) = \text{crown points}$ exactly, G is forced, on the region $\{\alpha : \text{pos}(\alpha) \in c_{an}(\text{Cr}_{an})\}$ (here c_{an} and c_{bn} are the crown embeddings $c_m : \text{Cr}_m \rightarrow [1]^m$ recalled at the head of this section, for $m = an, bn$), to take values in c_{bn} of the fibers $f_b^{-1}(f_a(i))$ — antichains of b points spaced $2n$ apart in Cr_{bn} , pairwise incomparable (Lemma 8.2). The region is the product of the crown subposet with the Boolean cube of negative literals, connected by Lemma 8.3; within each fiber of the crown projection the values lie in a single antichain, so by Lemma 8.4 G is constant in the negative directions, and its crown-region behaviour factors through a monotone $\rho : \text{Cr}_{an} \rightarrow \text{Cr}_{bn}$ with $f_b \rho = f_a$ — and $b \cdot \text{wind}(\rho) = a$ with $a < b$ has no integer solution. (Machine certificates: the crown-region constraint problems — 768 points for De Morgan, 312 for Kleene, both with domains of size $b = 4$ — are exhaustively unsatisfiable; §15.) For Kleene the same argument runs on the unmixed literal points, whose crown region remains connected (Lemma 8.3). $\square$

Remark 8.6. The Boolean cube category escapes: there the free algebras are determined by vertex actions, the set-level section *is* a cube map, the chain stabilizes immediately, and indeed the Boolean Karoubi envelope is the Eilenberg–Zilber category of finite nonempty sets [5].

The same phenomenon has an absolute form, which explains from first principles why elegance-based techniques require the *relative* theory over these sites:

Proposition 8.7. $\square_{\text{DM}}$ and $\square_{\text{KL}}$ are not Eilenberg–Zilber categories: the Kleene midpoint $x \wedge \neg x : [1] \rightarrow [1]$ factors through no (split epimorphism, monomorphism) pair.

Proof. The two points of $\mathbb{I}$ have the same image under $x \wedge \neg x$ (both go to $0 \wedge 1 = 0$), so the map is not a monomorphism; and it is not constant. Suppose $x \wedge \neg x = m \circ e$ with $e : [1] \rightarrow [k]$ a split epimorphism and $m : [k] \rightarrow [1]$ a monomorphism. The section of e exhibits $[k]$ as a retract of $[1]$, so the free algebra on k generators is a retract of the six-element free algebra on one generator; since $|\text{DM}(2)| = 168$ and $|\text{KL}(2)| = 84$ already exceed 6, this forces $k \leq 1$. If $k = 0$ the composite is constant — excluded. If $k = 1$: a split epimorphism $[1] \rightarrow [1]$ is given by an element t of the six-element free algebra admitting a compositional right inverse ($t \circ u = x$), and the only such elements are x and $\neg x$ (the constants and the connection-degenerate elements $x \wedge \neg x$, $x \vee \neg x$ satisfy $t \leq \neg t$ or $t \geq \neg t$, properties preserved under substitution and not satisfied by x); so e is invertible, and then $m = (x \wedge \neg x) \circ e^{-1}$ would be a monomorphism, contradicting the first sentence. $\square$

9. THE TEST COMPARISON RE-OPENED; THE UPPER BOUND

9.1. The folklore witness is void.

Remark 9.1. The 2018 record asserts that the cubical-type and test weak equivalences differ on De Morgan (and Boolean) cubes [11], with the reversal quotient as the underlying example. But L separates the cubical-type structure from the *topological realization*, not from the test structure: by Lemma 7.1 its test homotopy type is $B\mathbb{Z}/2$, which is precisely its cubical-type homotopy type (Corollary 5.5). On L the two structures *agree*, and it is the realization that is the outlier — it follows, incidentally, that the topological cocube on $\square_{\text{DM}}$ is not a test functor: it computes $*$ where the test structure computes $B\mathbb{Z}/2$. On the cartesian side the analogous question is genuinely settled in the negative — Coquand’s symmetric square is cubical-type-nontrivial [10] yet test-trivial (the equivariant structure contracts it and coincides with the test structure [1]) — but that argument turns on the fixed diagonal cells, which the reversal lacks. With the witness void and both repair routes closed (§§7–8), we know of no proof that $W_{\text{type}} \neq W_{\text{test}}$ on $\square_{\text{DM}}$, and we regard the question — *does CCHM cubical type theory present*

spaces through the test comparison? — as open. Proposition 2.1 gives a lower bound ($W_{\min} \subseteq W_{\text{type}}$); we now prove the upper bound.

9.2. The upper bound.

Theorem 9.2. *Let A be a totally aspherical test cube category (De Morgan, Kleene, Boolean, or cartesian), and suppose the trivial cofibrations of the cubical-type model structure on $\hat{A}$ are contained in the saturation — closure under pushout, transfinite composition, and retract — of the class $\{\delta_e \hat{\otimes} m : m \text{ a monomorphism, } e \in \{0, 1\}\}$. Then $W_{\text{type}} \subseteq W_{\text{test}}$.*

Proof. W_{test} is an A -localizer. (i) Since A is totally aspherical, el preserves finite products up to weak equivalence [9, §1.6], and $\text{el}(\mathbb{I})$ has a terminal object; hence the projection $X \times \mathbb{I} \rightarrow X$ is in W_{test} for every X , and by two-out-of-three so is each end inclusion $X \times \{e\} \hookrightarrow X \times \mathbb{I}$ — a monomorphism. (ii) For a monomorphism $m : X \rightarrowtail Y$, the map $\delta_e \hat{\otimes} m$ is the comparison $Y \cup_X (X \times \mathbb{I}) \rightarrow Y \times \mathbb{I}$. The inclusion $Y \hookrightarrow Y \cup_X (X \times \mathbb{I})$ is a pushout of the monomorphic test equivalence $X \hookrightarrow X \times \mathbb{I}$, hence in W_{test} by the localizer axioms; the end inclusion $Y \hookrightarrow Y \times \mathbb{I}$ is in W_{test} by (i); so $\delta_e \hat{\otimes} m \in W_{\text{test}}$ by two-out-of-three. (iii) Every map of the saturation is then a monomorphic member of W_{test} (the localizer axioms are closure properties of exactly this kind), so the trivial cofibrations of the type structure lie in W_{test} ; the trivial fibrations lie in every localizer; and an arbitrary map of W_{type} factors as a trivial cofibration followed by a trivial fibration. $\square$

Remark 9.3. The saturation hypothesis holds for the CCHM/Sattler construction, with no proper-class generation involved: the monomorphisms of a presheaf category are the saturation of the *set* of subrepresentable inclusions, the Leibniz product $\delta_e \hat{\otimes} (-)$ carries pushouts, transfinite composites and retracts to the same, so the fibrations — defined by uniform lifting against $\delta_e \hat{\otimes} m$ for all monomorphisms m — are equally detected by the *set* of $\delta_e \hat{\otimes} m$ with m subrepresentable, and the small object argument exhibits every trivial cofibration as a retract of a relative cell complex on these generators [16].

Corollary 9.4. *On cartesian cubical sets $W_{\text{type}} \subsetneq W_{\text{test}}$: the inclusion holds by Theorem 9.2, and it is strict because Coquand’s symmetric square is test-trivial [1] but not type-trivial [10].*

Remark 9.5. The contrapositive of Theorem 9.2 reads: a map that is *not* a test equivalence is *not* a type equivalence. All type-noncontractibility results below can be read off from element-category computations this way (Theorem 4.5 itself becomes a corollary of Lemma 7.1); we retain the syntactic proofs of §4 and §13, which are the only known technique working *inside* the type structure, and which independently guard the machinery.

10. QUOTIENTS THAT CONTRACT: THE FIXED-STRATUM RETRACTION

The remaining question is the lower bound, equivalently: is every test-trivial map type-trivial? A separating witness must be test-trivial and type-nontrivial. The candidates considered in this paper — which include all quotient witnesses documented in the discussions we are aware of — are isotropy quotients $\square^n/H$, $H \leq G_n$; in the remaining sections we compute both sides for all $H \leq G_2$ and a representative three-dimensional family, in the De Morgan theory.

Theorem 10.1. *Let $h \in G_n$ generate a cyclic group $C = \langle h \rangle$ and suppose h fixes at least one cell of $\square^n$. Then $\square^n/C \rightarrow 1 \in W_{\text{type}}$ (hence $\in W_{\text{test}}$).*

Proof. Write $h = (\pi, \varepsilon)$; by Lemma 3.2 the hypothesis means every cycle of π has even negation weight. Write σ for the substitution characterized by (generic cell) $\circ \sigma = h \cdot$ (generic cell). A homotopy $G = (G_1, \dots, G_n) : \square^n \times \mathbb{I} \rightarrow \square^n$ descends to $\square^n/C \times \mathbb{I} \rightarrow \square^n/C$ iff $G \circ \sigma = h \circ G$; componentwise,

$$G_i \circ \sigma = \neg^{\varepsilon_i} G_{\pi^{-1}(i)}.$$

Solving around a cycle. Fix a cycle $c = (i_1 \dots i_k)$ of π , of (even) weight w_c , and look for components G_{i_j} built from the cycle's variables $x_{i_1}, \dots, x_{i_k}$ and the homotopy variable t only. The equations determine $G_{i_2}, \dots, G_{i_k}$ from G_{i_1} (apply $\neg^\varepsilon$ and σ -transport along the cycle), and the chain closes with the single condition $G_{i_1} \circ \sigma^k = \neg^{w_c} G_{i_1}$. Now h^k is the identity on the coordinates of c — the cycle returns after k steps and the accumulated negation is $\neg^{w_c} = \text{id}$ — so the substitution σ^k is the identity *on the cycle's variables* (it may act nontrivially on the other cycles' variables, which is why we restricted G_{i_1} to the cycle's own variables); the closing condition is therefore vacuous, and G_{i_1} is free, subject only to its face conditions. These propagate around the cycle with the accumulated signs cancelling in pairs: $G|_{t=0} =$ the generic cell iff $G_{i_1}|_{t=0} = x_{i_1}$, and $G|_{t=1}$ is an h -fixed cell iff $G_{i_1}|_{t=1} = m_c$ with m_c compatible with the cycle transport, e.g. the cycle-symmetrized meet $m_c = \bigwedge_{j=1}^k \neg^{s_j} x_{i_j}$, where s_j is the accumulated weight from i_1 to i_j (well defined because w_c is even). Take the standard interpolation

$$G_{i_1} = (x_{i_1} \wedge \neg t) \vee (m_c \wedge t) \vee (x_{i_1} \wedge m_c).$$

Distinct cycles use disjoint variable sets and their equations do not interact, so the per-cycle solutions assemble into a descent-compatible homotopy from the identity to a retraction onto the fixed stratum.

The machine-verified instance. Consider the order-3 element $h(a, b, c) = (\neg c, \neg a, b)$ of G_3 — one 3-cycle of weight 2, whose fixed cells form the twisted diagonal $(a, \neg a, \neg a)$. The recipe gives $m = x \wedge \neg y \wedge \neg z$ and

$$G_1 = (x \wedge \neg t) \vee (m \wedge t) \vee (x \wedge m), \quad G_3 = \neg(G_1 \circ \sigma), \quad G_2 = \neg(G_1 \circ \sigma^2);$$

all identities — $\sigma^3 = \text{id}$, invariance of m , the three equivariance equations, the six face conditions, and monotonicity — are verified on all 256 points of the literal cube of $\text{DM}(x, y, z, t)$ (§15).

Conclusion. The homotopy contracts $\square^n/C$ onto its fixed stratum, a copy of $\square^{c(\pi)}$ ($c(\pi)$ the number of cycles; its cells are fixed, so the quotient identifies nothing there), which contracts to a vertex by the $\vee$ -connection within the stratum. $\square$

Corollary 10.2. $\square^2/\langle \text{sw} \rangle$ and $\square^2/\langle \text{g} \rangle$ are contractible in both structures: for sw take the invariant term $m = x \wedge y$ (recovering the connection contraction $q(s, t) \mapsto q(s \vee z, t \vee z)$), for g take $m = x \wedge \neg y$.

Remark 10.3. This is exactly what fails for cartesian cubes: Coquand’s square is $\square^2/\langle \text{sw} \rangle$ over a site with no connections, so no interpolation G_1 exists and the quotient is a genuine type-theoretic obstruction [10]. Over De Morgan cubes the connections dissolve every *cyclic* fixed-stratum obstruction; what survives is the mixed case of §12, where no cell is fixed by the whole group and no retraction target exists.

11. CELL-FREE QUOTIENTS

By Lemma 3.2, the cell-free subgroups of G_2 are those all of whose nontrivial elements have a cycle of odd negation weight: $\langle \text{n} \rangle$, $\langle \text{n}_1 \rangle$, $\langle \text{n}_2 \rangle$, $\langle \text{n}_1 \text{sw} \rangle \cong \mathbb{Z}/4$, and $\langle \text{n}_1, \text{n}_2 \rangle$.

Theorem 11.1. *If $H \leq G_n$ is cell-free then $\square^n/H$ is a $K(H, 1)$ in the cubical-type model structure, and $\text{Nel}(\square^n/H) \simeq BH$: the two structures agree on the nose on all cell-free quotients.*

Proof. Theorem 5.4 and Lemma 7.1. $\square$

12. THE MIXED QUOTIENTS: THE JOIN

Two subgroups of G_2 remain: the Klein group K and G_2 itself. No cell of $\square^2$ is fixed by all of K : a fixed cell would satisfy $b = a$ (from sw) and $a = \neg a$ (from n), which is impossible. The stabilizer of a cell (a, b) is $\langle \text{sw} \rangle$ if $b = a$ (the *diagonal stratum*), $\langle \text{g} \rangle$ if $b = \neg a$ (the *antidiagonal stratum*), and trivial otherwise (the *free stratum*).

We will use a standard decomposition of the nerve of a *collage*:

Lemma 12.1 (nerve of a collage). *Let X be a category partitioned into a full subcategory S that receives no morphisms from its complement (a sieve) and the complementary full subcategory U (a cosieve), and let $\widehat{W}$ be the category whose objects are the morphisms $s \rightarrow u$ ($s \in S, u \in U$) and whose morphisms are the commuting squares. Then NX is weakly equivalent to the double mapping cylinder $\text{hocolim}(NS \leftarrow N\widehat{W} \rightarrow NU)$ along the two projections.*

Proof. Write $W(s, u) := X(s, u)$ for the hom profunctor.

Step 1 (crossing decomposition). Since no morphism leads from U into S , every r -simplex of NX is either a chain in S , a chain in U , or decomposes uniquely as $s_0 \rightarrow \cdots \rightarrow s_p \xrightarrow{\omega} u_0 \rightarrow \cdots \rightarrow u_q$ with $p + q = r - 1$ and $\omega \in W(s_p, u_0)$ (the remaining mixed arrows of the simplex are the composites, determined by this data). Hence

$$(NX)_r = (NS)_r \sqcup (NU)_r \sqcup \bigsqcup_{p+q=r-1} Y_{p,q}, \quad Y_{p,q} := \bigsqcup_{\sigma, \tau} W(s_p, u_0),$$

the sum over p -chains σ in S and q -chains τ in U ; the seam faces act by the two profunctor actions, the interior faces within σ and τ .

Step 2 (the middle computes $N\widehat{W}$). For a fixed p -chain σ , the simplicial set $q \mapsto \bigsqcup_{\tau} W(s_p, u_0)$ is the nerve of the category $s_p \backslash W$ of pairs $(u, \omega \in W(s_p, u))$. Thus Y , read as the simplicial object $[p] \mapsto \bigsqcup_{\sigma} N(s_p \backslash W)$, is the simplicial replacement of the diagram $s \mapsto N(s \backslash W)$, so its realization computes its homotopy colimit [3, XII.5.2], which by Thomason's theorem [17, Thm. 1.2] is the nerve of the Grothendieck construction of $s \mapsto s \backslash W$ — and that Grothendieck construction is exactly $\widehat{W}$. Under this equivalence the two augmentations of Y (forget (ω, τ) ; forget (σ, ω)) become the projections $N\widehat{W} \rightarrow NS$ and $N\widehat{W} \rightarrow NU$.

Step 3 (assembly). Filter NX by $NS \sqcup NU$ together with the nondegenerate crossing simplices attached in order of dimension, and filter the double mapping cylinder correspondingly by the cells of Y . Attaching a nondegenerate crossing r -simplex of type (p, q) attaches, on the cylinder side, the corresponding (p, q) -cell of Y crossed with Δ^1 , its ends attached to the σ -cell of NS and the τ -cell of NU ; on both sides the attachment is along the same decomposition of the boundary (σ -faces and τ -faces matching the cylinder ends, seam faces matching ∂ of the Y -cell), and the two attached cells are compared by the classical equivalence between the join $\Delta^p \star \Delta^q \cong \Delta^{p+q+1}$ and the double mapping cylinder of the two projections of $\Delta^p \times \Delta^q$ — a homotopy equivalence relative to the common boundary decomposition. The gluing lemma propagates the comparison over each attachment and the colimit, giving $NX \simeq \operatorname{hocolim}(NS \leftarrow N\widehat{W} \rightarrow NU)$. (Sanity check: for $S = U = *$ and W a two-element set, both sides give the circle.) $\square$

Theorem 12.2. $\operatorname{el}(\square^2/K) \simeq B\mathbb{Z}/2 \star B\mathbb{Z}/2$, the join of two infinite projective spaces: simply connected, with $\widetilde{H}_2 = 0$ and $H_3 = \mathbb{Z}/2$. Moreover $\operatorname{el}(\square^2/G_2) \simeq \operatorname{hocolim}(B\mathbb{Z}/2 \leftarrow BK \rightarrow BG_2)$, which has fundamental group $\mathbb{Z}/2$.

Proof. Write $C = \operatorname{el}(\square^2)$, the slice $\square_{\text{DM}}/[2]$, a contractible category (terminal object $T = \text{id}$) on which K acts by postcomposition; as in Lemma 7.1, the category of elements of the quotient presheaf is the strict quotient: $\operatorname{el}(\square^2/K) \cong C/K$.

Step 1 (sieve structure). Along any morphism $[f] \rightarrow [g]$ of C/K one has $\operatorname{Stab}(g) \subseteq \operatorname{Stab}(f)$: if $g \circ u = f$ and $k \in \operatorname{Stab}(g)$ then $kf = kgu = gu = f$

(the group is abelian, so stabilizers are well-defined on orbits). There are no morphisms from the free stratum into a stabilized one, and none between the two stabilized strata: a morphism into the diagonal stratum forces equal components on a K -orbit that has none (verified also by machine over all 23,040 small-level composites, zero violations; §15). So the two stabilized strata form a sieve $S = D \sqcup A$ and the free stratum a cosieve U , and Lemma 12.1 applies.

Step 2 (the three pieces).

- U : the free stratum contains T and is closed under maps out, so it is contractible before quotienting; the K -action on it is free, so $U \simeq BK$ as in Lemma 7.1.
- D : the diagonal stratum of C is $\text{el}(\mathbb{I}) \simeq *$ via $a \mapsto (a, a)$; sw acts trivially on it and $\mathfrak{n}$ freely, so $D \simeq B(K/\langle \text{sw} \rangle) = B\mathbb{Z}/2$ — a copy of $\text{el}(L)$. Likewise $A \simeq B(K/\langle \mathfrak{g} \rangle)$.
- $\widehat{W}_D$: it is isomorphic to $\text{Arr}(D_C, U_C)/K$, the quotient of the category of arrows from diagonal to free objects of C . The projection $\text{Arr} \rightarrow D_C$ is a Grothendieck fibration whose fiber over d is the comma $(d \downarrow U_C)$, which has terminal object the unique map $d \rightarrow T$; so Arr is contractible by Quillen's Theorem A, the K -action on it is free (its free-stratum component is), and $\widehat{W}_D \simeq BK$.

Step 3 (the maps, and the join). The functor $\widehat{W}_D \rightarrow U$ is the free quotient of an equivariant map of contractible categories, hence an equivalence $BK \rightarrow BK$; the functor $\widehat{W}_D \rightarrow D$ is equivariant along $K \twoheadrightarrow K/\langle \text{sw} \rangle$, hence the classifying map of that projection. Gluing the two cylinders along the equivalences leaves

$$\text{el}(\square^2/K) \simeq \text{hocolim}(B(K/\langle \text{sw} \rangle) \longleftarrow BK \longrightarrow B(K/\langle \mathfrak{g} \rangle)).$$

Since $K = \langle \text{sw} \rangle \times \langle \mathfrak{g} \rangle$, the two legs are the projections $B\mathbb{Z}/2 \times B\mathbb{Z}/2 \rightarrow B\mathbb{Z}/2$, and the homotopy pushout of the two projections of a product is by definition the join. Joins of path-connected spaces are simply connected, and $\widetilde{H}_n(X \star Y) \cong \widetilde{H}_{n-1}(X \wedge Y)$ gives $\widetilde{H}_2 = 0$ and $H_3 = H_1 \otimes H_1 = \mathbb{Z}/2$.

For G_2 the reflection $\mathfrak{n}_1$ interchanges the two stabilized strata, which merge into one with $S \simeq B(\langle \text{sw}, \mathfrak{n} \rangle / \langle \text{sw} \rangle)$, and the interface becomes BK mapping to $U \simeq BG_2$ along the inclusion $K \hookrightarrow G_2$; van Kampen gives $\pi_1 = G_2 / \langle \langle \text{sw} \rangle \rangle = G_2 / K = \mathbb{Z}/2$ (the normal closure of sw contains its $\mathfrak{n}_1$ -conjugate $\mathfrak{g}$). $\square$

Corollary 12.3. $\square^2/K \rightarrow 1 \notin \mathcal{W}_{\text{type}}$ and $\square^2/G_2 \rightarrow 1 \notin \mathcal{W}_{\text{type}}$: both quotients are contractible in neither structure.

Proof. Immediate from Theorem 12.2 and the contrapositive of Theorem 9.2. $\square$

13. THE SYNTACTIC CENSUS FOR THE KLEIN QUOTIENT

Corollary 12.3 passes through the test structure. We record an independent proof of $\square^2/K \rightarrow 1 \notin \mathbf{W}_{\text{type}}$ that never leaves the type structure, by the method of §4; it doubles as a verification of the census machinery.

Call a square c of a De Morgan cubical set K -*symmetric* if $c \circ \sigma_{\text{sw}} = c$ and $c \circ \sigma_{\text{n}} = c$, where $\sigma_{\text{sw}}, \sigma_{\text{n}}$ are the corresponding automorphisms of $[2]$ acting by precomposition (for cells of a quotient this is orbit equality, so the condition unfolds to a twisted equation with a twist in K); similarly for cubes, with the extra direction untouched. Let $P''(X)$ be the property: every K -symmetric square of X is connected to a constant square by a finite chain of K -symmetric cubes.

Proposition 13.1. *$P''(\square^2/K)$ fails: the K -symmetric squares number 72 and the chain graph decomposes into the component of the generic square (size 3), the component of the constants (size 66), and one further component (size 3); the generic square reaches no constant.*

The census enumerates, for each assignment of a twist in K to each of the two generators, the compatible cubes: a twist with a transposition component determines one component of the cube from the other over all of $\text{DM}(3)$, and the purely diagonal twists reduce to face-profile combinations; positive and negative controls (the cyclic subgroups, which must connect by Corollary 10.2, and the cell-free subgroups, which must not, by Theorem 11.1) all behave as theory predicts (§15). That 66 of 72 symmetric squares *do* contract shows the invariant is nondegenerate: the isolation of the generic square is not a formal artifact.

Proposition 13.2. *P'' propagates through the syntactic fibrant replacement: for the factorization $1 \rightarrow T \twoheadrightarrow \square^2/K$, $P''(1)$ implies $P''(T)$; and if $u : X \rightarrow Y$ has a section, $P''(X)$ implies $P''(Y)$. Consequently $\square^2/K \rightarrow 1 \notin \mathbf{W}_{\text{type}}$.*

Proof. As in Proposition 4.3, with one new ingredient: the substitutions $\sigma_{\text{sw}}, \sigma_{\text{n}}$ are automorphisms of the free De Morgan algebra on two generators, so no restriction clause of a **comp**-cell can fire under them, symmetry of a composition cell is an equality of **comp**-constructors, and all its data (directions, constraint, tube, base) is itself symmetric. The fresh-direction cube $c(x, y, t) = (v(x, y, t), \text{comp}^{k \rightarrow t}(a, \psi, u, \langle z \rangle v(x, y, z)))$ is then K -symmetric, agrees with the given square at $t = l$, and reduces the inductive height at $t = k$; the induction terminates at cells of the point. Given $\square^2/K \rightarrow 1 \in \mathbf{W}_{\text{type}}$, two-out-of-three makes the vertex $1 \rightarrow \square^2/K$ a weak equivalence, the fibration $T \rightarrow \square^2/K$ is then trivial and has a section, and $P''(1) \Rightarrow P''(T) \Rightarrow P''(\square^2/K)$ contradicts Proposition 13.1. $\square$

14. AGREEMENT, NON-SEPARATION, AND THE CONJECTURE

Corollary 14.1. *No quotient $\square^2/H$, $H \leq G_2$, is a separating witness for $\mathbf{W}_{\text{type}} \subseteq \mathbf{W}_{\text{test}}$ on De Morgan cubical sets: the subgroups of G_2 are exhausted*

quotient	cubical-type structure	test structure
$L = \mathbb{I}/\langle \neg \rangle$	$K(\mathbb{Z}/2, 1)$ (Cor. 5.5)	$B\mathbb{Z}/2$ (Lem. 7.1)
$\square^2/\langle \text{sw} \rangle, \square^2/\langle \text{g} \rangle$	$*$ (Cor. 10.2)	$*$ (Thm. 9.2)
$\square^3/\mathbb{Z}/3$ twisted	$*$ (Thm. 10.1)	$*$ (Thm. 9.2)
cell-free H	$K(H, 1)$ (Thm. 5.4)	BH (Lem. 7.1)
$\square^2/K$	$\neq *$ (Cor. 12.3, Prop. 13.2)	$\mathbb{RP}^\infty \star \mathbb{RP}^\infty$ (Thm. 12.2)
$\square^2/G_2$	$\neq *$ (Cor. 12.3)	$\pi_1 = \mathbb{Z}/2$ (Thm. 12.2)

TABLE 1. No quotient analyzed here separates the two localizers: the weak-contractibility verdicts agree in every case, and on the reversal circle and the cell-free quotients the full homotopy types agree. (For the mixed quotients the cubical-type column records non-contractibility; the full cubical-type homotopy type there is the subject of Conjecture 14.2.)

by Corollary 10.2 (cyclic with fixed cells), Theorem 11.1 (cell-free), and Corollary 12.3 (mixed), and in every case the quotient is test-trivial if and only if it is type-trivial.

Conjecture 14.2. (a) *The homotopy types agree on all isotropy quotients in all dimensions: for every $H \leq G_n$, the cubical-type structure assigns to $\square^n/H$ the homotopy colimit of its stabilizer strata — the same diagram that computes $\text{el}(\square^n/H)$.*

(b) $W_{\text{type}} = W_{\text{test}}$ on De Morgan cubical sets.

Part (a) is evidence for part (b), and every result of this paper is consistent with both; but (b) does not follow from (a) alone: one would additionally need a generation theorem exhibiting W_{test} as the localizer generated by maps covered by (a) — see direction (2) of the outlook below. Theorem 8.5 rules out the elegance-based cellular routes to such a generation statement, which is precisely why we state (b) as a conjecture rather than as a consequence of (a).

Remark 14.3. Two remarks on why the remaining inclusion resists. First, the geometric realization of every quotient above is contractible (a folded square is a triangle), so no cellular or realization argument can see the answer; the phenomenon lives strictly in the difference between strict and homotopy quotients, on both sides. Second, the element-category computation of Theorem 12.2 rests on a stratification that has *no presheaf-level counterpart*: the subpresheaf of $\square^2$ generated by the free cells is all of $\square^2$, because every diagonal cell (a, a) is a face of the free cell $(a, a \vee (t \wedge \neg t))$ — a Kleene-midpoint phenomenon with no Boolean or cartesian analogue. A type-side computation of the mixed quotients therefore needs a genuinely new device (a presheaf-level collage), which we leave open.

Outlook. The results funnel toward one question, Conjecture 14.2. Three concrete directions. (1) *The type side of the join*: a presheaf-level double

mapping cylinder interpolating between $\square^2/K$ and the join, with the two L -copies (Corollary 5.5) as the stabilized ends. (2) *Generation of W_{test}* : the equality $W_{\text{type}} = W_{\text{test}}$ would follow from exhibiting a generating set of the localizer W_{test} consisting of maps already in W_{type} ; over a totally aspherical site the natural candidates are exactly the quotient maps studied here, whose membership in W_{type} is what part (a) of Conjecture 14.2 would supply — this generation statement is the missing step between parts (a) and (b). (3) *Beyond quotients*: Corollary 14.1 closes the two-dimensional quotient route to a counterexample; a separating witness, if any, must be either higher-dimensional with non-abelian mixed isotropy, or not an isotropy quotient at all. Independently, the equivariant model structure outlined in Remark 7.4 would present *some* homotopy theory — a localization of the test theory in which the classifying spaces BH of cell-free subgroups are contracted — whose identification we leave open; and the orbit-groupoid form of the vertex lemma may be of use wherever the machinery of [1] meets non-vertex-fixing symmetries.

15. MACHINE VERIFICATION

All finite combinatorial claims were verified by exhaustive enumeration (not by a proof assistant: the verifications are reproducible computations, with the mathematical proofs given in the text). The scripts accompany the artifact [21] and are independently reproducible from the pinned release.

Freeness (Theorem 3.1): 0 self-dual elements in $\text{DM}(m)$ for $m \leq 3$ (out of 6, 168, 7,828,354) and in $\text{KL}(m)$ for $m \leq 3$ (out of 6, 84, 43,918).

Reversal census (Proposition 4.4): the symmetric squares number $60 / 32 / 8$ (De Morgan / Kleene / Boolean); their face graphs on symmetric lines have edge counts $[x] \rightarrow [x]$: $10/6/4$, degenerate $\leftrightarrow$ constant: $10+10 / 6+6 / \text{---}$, constant $\rightarrow$ constant: $10/8/4$, and *no* edge leaves the $[x]$ -component in any theory. No rescue terms: no t in $\text{DM}(x, z)$ or $\text{KL}(x, z)$ is reversal-compatible, restricts to the generator at $z = 0$, and is constant at $z = 1$.

Automorphisms (Theorem 6.1): $8 = |G_2|$ coordinate permutations commuting with the involution for De Morgan at $n = 2$. For Kleene at $n = 2$: the unmixed poset has $14 = 2 \cdot 3^2 - 2^2$ points; the monotone Boolean functions on it number $84 = |\text{KL}(2)|$, confirming the representation; its poset automorphisms number exactly 8, all commuting with the involution; and the same-pair atom criterion of the proof is confirmed — the two atoms of one pair have two minimal upper bounds, atoms of distinct pairs a least upper bound. The non-commutation $(\neg x) \wedge e \neq \neg(x \wedge e)$ shows hyperoctahedral equivariance is not implied by connections (the connection-contraction argument for automatic symmetric-group equivariance [6, Rem. 4.25] commutes with permutations only).

Crown regions (Theorem 8.5): for $(n, a, b) = (3, 2, 4)$, the De Morgan region has 768 points with 16,728 order constraints and the Kleene region 312

points, all domains of size 4; backtracking with propagation finds no solution in either, while the fiber-incomparability and three-branch-disjointness lemmas are checked directly.

Isotropy censuses (§13): over the subgroups of G_2 , with nodes the H -symmetric squares and edges the H -symmetric cubes: $\langle \mathbf{sw} \rangle$ and $\langle \mathbf{g} \rangle$: 476 nodes, 173,180 edges, generic square connected to a constant (as Corollary 10.2 requires); $\langle \mathbf{n} \rangle$: 464 nodes, not connected (component 72); $\langle \mathbf{n}_1 \rangle$: 1,500 nodes, not connected (component 250) — as Theorem 11.1 requires; K : 72 nodes, 2,094 edges, components $\{3, 66, 3\}$, generic isolated from the constants; G_2 : 18 nodes, 134 edges, generic isolated. *Sieve structure* (Theorem 12.2): 23,040 small-level composites from free to stabilized objects, zero admissible. *Fixed-stratum retraction* (Theorem 10.1): all identities on all 256 literal points of $\mathrm{DM}(x, y, z, t)$.

Artifact. The verification scripts and the machine-checked companions are available in the artifact repository:

<https://github.com/r-shibusawa/cubical-strict-j>,
release v1.12.0, commit f15e4864,

archived at the version DOI 10.5281/zenodo.21785412 (concept DOI for all versions: 10.5281/zenodo.21405961). The repository hosts the machine-checked companion series to which [19, 20] belong; the scripts for the present paper are `reversal.py`, `equivariance.py`, `nogo.py` and `isotropy.py`.

REFERENCES

- [1] S. Awodey, E. Cavallo, T. Coquand, E. Riehl, C. Sattler. The equivariant model structure on cartesian cubical sets. *Adv. Math.* 495 (2026), 110965.
- [2] J. Bourke, R. Garner. Algebraic weak factorisation systems I: accessible AWFS. *J. Pure Appl. Algebra* 220 (2016), 108–147.
- [3] A. K. Bousfield, D. M. Kan. *Homotopy Limits, Completions and Localizations*. Lecture Notes in Math. 304, Springer, 1972.
- [4] U. Buchholtz, E. Morehouse. Varieties of cubical sets. *RAMiCS 2017*, LNCS 10226, 77–92, 2017.
- [5] T. Champion. Cubical sites as Eilenberg–Zilber categories. Preprint, arXiv:2303.06206, 2023.
- [6] E. Cavallo, C. Sattler. Relative elegance and cartesian cubes with one connection. *Canad. J. Math.*, published online 2025, doi:10.4153/S0008414X25101466; arXiv:2211.14801.
- [7] E. Cavallo, C. Sattler. Eliminating reversals from cubical type theories. *LICS 2026*, LIPIcs 380, 2026.
- [8] C. Cohen, T. Coquand, S. Huber, A. Mörtberg. Cubical type theory: a constructive interpretation of the univalence axiom. *TYPES 2015*, LIPIcs 69, 5:1–5:34, 2018.
- [9] D.-C. Cisinski. *Les préfaisceaux comme modèles des types d’homotopie*. Astérisque 308, Soc. Math. France, 2006.
- [10] T. Coquand. Trivial cofibration-fibration factorization with one application. Unpublished note, June 2018; circulated on the Homotopy Type Theory mailing list [11].
- [11] T. Coquand et al. Quillen model structure. Homotopy Type Theory mailing-list discussion, June 2018. https://groups.google.com/g/homotopytypetheory/c/RQkLWZ_83kQ.
- [12] M. Hovey. *Model Categories*. Math. Surveys Monogr. 63, AMS, 1999.

- [13] G. Maltsiniotis. La catégorie cubique avec connexions est une catégorie test stricte. *Homology Homotopy Appl.* 11 (2009), 309–326.
- [14] Y. Movsisyan, V. Aslanyan. De Morgan functions and free De Morgan algebras. *Demonstr. Math.* 47 (2014), 271–283.
- [15] C. Sattler. Do cubical models of type theory also model homotopy types? Lecture, Hausdorff Trimester Program “Types, Sets and Constructions”, Bonn, June 2018.
- [16] C. Sattler. The equivalence extension property and model structures. Preprint, arXiv:1704.06911, 2017.
- [17] R. W. Thomason. Homotopy colimits in the category of small categories. *Math. Proc. Cambridge Philos. Soc.* 85 (1979), 91–109.
- [18] T. Streicher, J. Weinberger. Simplicial sets inside cubical sets. *Theory Appl. Categ.* 37 (2021), 276–286.
- [19] R. Shibusawa. Boundary fibers in free De Morgan algebras: Boolean intervals, Sperner bounds, and median contractions. Preprint, 2026. In the artifact repository (release v1.6.0, DOI 10.5281/zenodo.21695307).
- [20] R. Shibusawa. The Kleene interpolation: boundary fibers and the strict layer across the subvarieties of De Morgan algebras. Preprint, 2026. In the artifact repository (release v1.8.1, DOI 10.5281/zenodo.21695917).
- [21] R. Shibusawa. Cubical strict-J artifact. <https://github.com/r-shibusawa/cubical-strict-j>, release v1.12.0, DOI 10.5281/zenodo.21785412.

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